



Multiplying with decimals

Task A: Multiplying with decimals

1) Work out the following:

a) $8.3 \times 11 =$

×	80	3
10	800	30
1	80	3

$$\begin{aligned} 8.3 \times 11 &= \\ 83 \times 0.1 \times 11 &= \\ 83 \times 11 \times 0.1 &= \\ 913 \times 0.1 &= 91.3 \end{aligned}$$

b) $3.7 \times 6.4 =$

×	30	7
60	1800	420
4	120	28

$$\begin{aligned} 3.7 \times 6.4 &= \\ 37 \times 0.1 \times 64 \times 0.1 &= \\ 37 \times 64 \times 0.01 &= \\ 2368 \times 0.01 &= 23.68 \end{aligned}$$

c) 2.1×1.28

×	20	1
100	2000	100
20	400	20
8	160	8

$$\begin{aligned} 21 \times 0.1 \times 128 \times 0.01 &= \\ 21 \times 128 \times 0.001 &= \\ 2688 \times 0.001 &= 2.688 \end{aligned}$$

2) Tick the correct calculations.

A: $52.4 \times 1.6 = 524 \times 16 \div 100$ ✓

B: $52.4 \times 1.6 = 524 \times 16 \div 10$

C: $52.4 \times 1.6 = 524 \times 16 \times 0.01$ ✓

D: $5.24 \times 1.6 = 524 \times 16 \div 1000$ ✓

E: $0.524 \times 1.6 = 524 \times 16 \div 1000$

3) Given $456 \times 12 = 5472$

Work out:

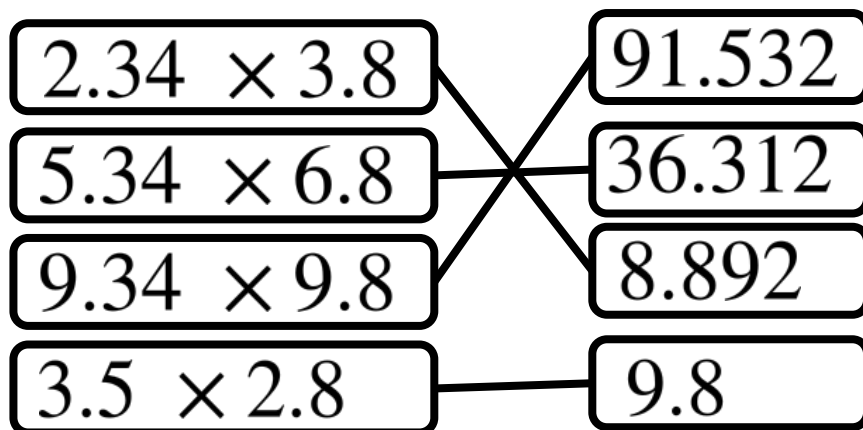
a) $45.6 \times 12 = 547.2$ We know $456 \times 12 = 5472$ so $456 \times 0.1 \times 12 = 547.2$

b) $4.56 \times 12 = 54.72$ We know $456 \times 12 = 5472$ so $456 \times 0.01 \times 12 = 54.72$

c) $4.56 \times 1.2 = 5.472$ We know $456 \times 12 = 5472$ so $456 \times 0.01 \times 12 \times 0.1 = 5.472$

Task B: Appropriate answers to calculations

- 1) Using your knowledge of identifying a reasonable answer and without working out, match the calculation to the correct answer.



- 2) Are the statements always true, sometimes true or never true? You can use a calculator to help justify your answers.

Statement	Always true	Sometimes true	Never true	Justify
The product of two decimals is always a decimal.		✓		E.g. $1.5 \times 2.3 = 3.45$ E.g. $2.5 \times 4.4 = 11$
The product of two decimals each with 2 decimal places has 4 decimal places.		✓		E.g. $1.36 \times 1.25 = 1.7$ E.g. $0.45 \times 0.89 = 0.4005$
The product of two decimals is always smaller than the largest multiplier.		✓		E.g. $4.6 \times 1.3 = 5.98$ E.g. $34.1 \times 0.4 = 13.64$

